

# Can a Paper's Title Predict Its Citations?

Sorting computer-science papers into citation bands, and finding out which information actually helps

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Problem Statement 66

**43,997**

CS papers from OpenAlex,  
published 2019 to 2023

**0.105**

macro-F1 of always guessing  
the biggest band

**0.361**

macro-F1 from the  
title alone

**0.450**

macro-F1 with title, metadata  
and venue together

## 1 The problem

When a computer-science paper comes out, what do we know that says how often it will be cited? We sort each paper into one of four bands, from no citations at all to eight or more in its first two years. It is a plain multi-class classification task. It has nothing to do with citation networks or with why one paper cites another, even though the problem title can be read that way.

We started from a CS229 project (Fall 2019) that predicted citation bins from a paper's title, journal and year. Its authors saw scores stop improving however large the network got, and suggested that the input features were the limit, but they did not test that. That became our question: if a bigger model does not help, is it the information we feed it? We also changed four things in the setup. The original predicted raw citation counts, which depend on how long a paper has been out; we count citations inside a fixed window. Its bin edges were taken from the training data; ours are fixed and explained. Its report compares models with each other; ours also shows a majority-class baseline next to every score. And its journal was encoded as one numeric code; ours is one-hot encoded and tested on its own.

## 2 The data

No dataset came with the problem, so we built one from the OpenAlex API. We took English-language works of OpenAlex type 'article' in Computer Science from 2019 to 2023 that have a title and at least one reference, and drew 800 papers at random for every year and CS subfield (5 years  $\times$  11 subfields = 55 groups, fixed seed 42). That gives **43,997 papers**, each with its title, year, subfield, topic, venue, number of references and citation counts. The raw citation total is kept for comparison but never used as a feature. Only the venue is ever missing (602 papers, 1.4%). Papers from 2024 are left out because their second citation year is not finished.

The data lives on Kaggle ([tharunganesh172/citations-openalex-cs-2019-2023](https://www.kaggle.com/tharunganesh172/citations-openalex-cs-2019-2023)). We split it once, 70/15/15 into train, validation and test (30,797 / 6,600 / 6,600 papers, balanced by band and year), and saved the split in `data/split.csv` so that every experiment sees exactly the same papers.

## 3 The target and the four bands

Total citations favour old papers, so we count only the citations a paper receives in the two years after the year it was published. Every paper is then measured over the same two-year window.

The counts are very uneven: 26.5% of papers get none, the median is 2 and the 99th percentile is 64. We cut them into four bands at about the quartiles: **0, 1 to 2, 3 to 7, and 8 or more** citations, called Uncited, Low, Medium and High. Each band holds roughly a quarter of the papers, so always guessing the biggest band scores only 0.265 accuracy and 0.105 macro-F1, and random guessing scores 0.25.

Figure 1 is our check that the window works. If it does, papers from different years should look alike, and they do: no trend in either number.

## 4 How we tested what matters

We keep the model simple and change what it is allowed to see. Every score in this report comes from multinomial logistic regression. The features come from one shared pipeline with three blocks that can be switched on and off: the **title** (word and character TF-IDF), the **metadata** (subfield, topic, year and the log of the number of references) and the **venue** name. Rare venues and topics are grouped together.

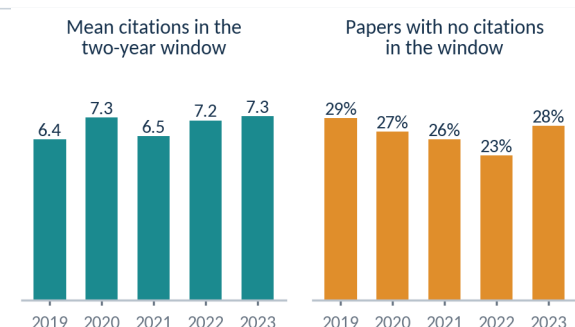


Figure 1. Papers from every year look alike, so paper age no longer drives the target.

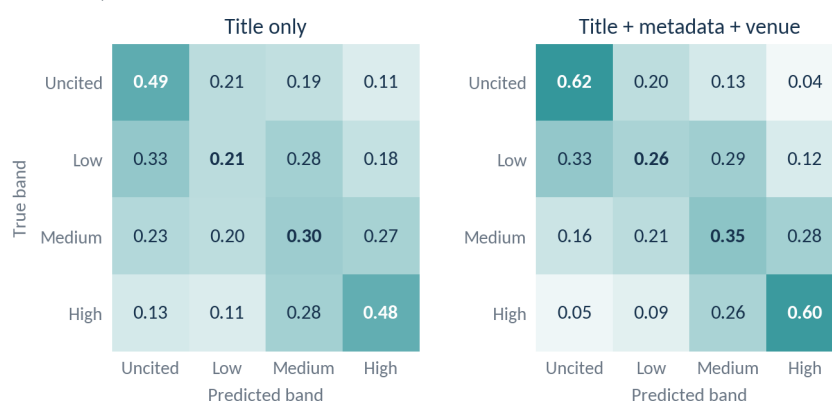
Each combination was tested with 5-fold cross-validation on the 37,397 training and validation papers, then scored **once** on the 6,600 test papers. Our main score is macro-F1, the average of the four per-band F1 scores, which cannot be raised by always predicting the biggest band. We used the same split, the same metric code and one shared results log (the RandomForest and HistGradientBoosting runs sit in a separate file), and the code is covered by automated tests. A small web demo takes a title and shows the predicted band, the probabilities and the words that pushed the prediction. It only sees a title, so it runs the title-only model.

## 5 What we found

| What the model sees             | 5-fold CV    | Test macro-F1 | Test accuracy |
|---------------------------------|--------------|---------------|---------------|
| Always guess the biggest band   | n/a          | 0.105         | 0.265         |
| Title only                      | 0.372        | 0.361         | 0.368         |
| Venue only                      | 0.385        | 0.390         | 0.420         |
| Title + metadata                | 0.424        | 0.420         | 0.429         |
| Metadata only                   | 0.427        | 0.423         | 0.443         |
| <b>Title + metadata + venue</b> | <b>0.448</b> | <b>0.450</b>  | <b>0.459</b>  |

**Table 1.** Macro-F1 from cross-validation and on the held-out test set. The two agree to within about 0.01 (the largest gap is 0.011, for the title alone), so nothing was tuned on the test set.

- The title helps, but only a little.** It beats the baseline clearly (0.361 against 0.105), yet it is the weakest source of information. Once the metadata is known it adds nothing: 0.423 without the title, 0.420 with it.
- Metadata and venue say more.** In our first notebook runs the number of references alone gave about 0.40, and the venue alone gives 0.39. Adding the venue to title and metadata lifts the score from 0.420 to 0.450.
- A bigger model does not fix it.** On the validation split, Naive Bayes and a linear SVM on titles stayed within 0.01 of logistic regression, and RandomForest and HistGradientBoosting did not beat it either (metadata only: 0.429 for logistic regression, 0.403 for RandomForest, 0.422 for HistGradientBoosting). The ceiling of about 0.45 comes from the features, which is what the original project suspected.
- The middle is the hard part.** With everything switched on, 62% of uncited papers and 60% of papers with 8 or more citations land in the right band, and the worst mistakes (an uncited paper called High, or the reverse) drop to 4 to 5%, down from 11 to 13% with the title alone (Figure 2). The Low band, 1 to 2 citations, is hard in every setup (test F1 between 0.20 and 0.29).



**Figure 2.** Test-set confusion matrices. Each row shows where papers of one true band end up, so the diagonal is correct and the corners are the worst mistakes.

## 6 What it adds up to

A paper's title says little about how often it will be cited. Its venue and the length of its reference list say more, and the best we can do from what is known at publication is about 0.45 macro-F1. That is well above the baseline but far from precise, and the demo shows it: a title alone gets 0.36.

**Limits.** We used a sample of 44,000 papers, not the whole field. Our main comparison uses one model family; RandomForest and HistGradientBoosting were run on the validation split only, and HistGradientBoosting only on venue and metadata because the title features were too large to convert. We did not try neural models. The reference count probably stands in for the kind of paper (surveys cite a lot), so it should not be read as a cause of citations. The venue column has 9,658 distinct values (7,870 in the training split), which limits what simple one-hot encoding can learn. Abstracts, author information and venue embeddings are the obvious next steps.

Code: [github.com/Tharun-2605/citation-classification](https://github.com/Tharun-2605/citation-classification) | Data: [kaggle.com/datasets/tharunganesh172/citations-openalex-cs-2019-2023](https://kaggle.com/datasets/tharunganesh172/citations-openalex-cs-2019-2023)